



ضع دائرة حول اسم مدرستك: (1) سليمان بني أحمد، (2) وفاء ضبابات، (3) كرم غنيمات، (4) اسماء الختوم، (5) ديانا طهراوي

1B	2	3	4	5

6	7	8	9	10

11	12	13	14	15

16	17	18	19	20

Question 1: Multiple choice questions (20 points)

1. The output of a logic gate can be one of two ____.

- a. Inputs
- b. Gates
- c. States
- d. none

2. Full adder is constructed by using

- a. Two Half Adder & one OR gate
- b. Two OR gate & one HA
- c. One HA & two OR gate
- d. One OR gate & one HA

3. A half adder gives..... Output(s)

- a. 1
- b. 2
- c. 3
- d. none of the above

4. A full adder gives..... Output(s)

- a. 1
- b. 2
- c. 3
- d. non of this

5. The output of Half adder is in the form of.

- a. Sum
- b. carry
- c. sum & carry
- d. none of these

6. Logic 0 is represented by GND practically

- a. True
- b. False

7. $A'B' = (AB)'$

- a. True
- b. False

8. $AB = (A' + B')'$

- a. True
- b. False

9. The red light on the LED indicates to result = 0.

- a. True
- b. False

10. The output of an AND gate with three inputs, A, B, and C, is HIGH when

- a. A = 1, B = 1, C = 0
- b. A = 0, B = 0, C = 0
- c. A = 1, B = 1, C = 1
- d. A = 1, B = 0, C = 1

11. The Boolean expression for a 3-input AND gate is

- a. $X = AB$
- b. $X = ABC$
- c. $X = A + B + C$
- d. $X = AB + C$

12. Logic states can only be ____ or 0.

- a. 3
- b. 2
- c. 1
- d. 0

13. The output of a ____ gate is only 1 when all of its inputs are 1.

- a. NOR
- b. XOR
- c. AND
- d. NOT

14. A NAND gate is equivalent to an AND gate plus a gate put together.

- a. NOR
- b. NOT
- c. XOR
- d. none

15. Half adder circuit is ____.

- a. Half of an AND gate
- b. A circuit to add two bits together
- c. Half of a NAND gate
- d. none of above

16. Which logic gate does the truth table shown in figure 1 have?

- a. A two-input AND gate.
- b. A two-input OR gate.
- c. An exclusive OR gate.
- d. An exclusive NOR gate.

17. What Boolean expression describes the output X of the arrangement shown in figure 2?

- a. $X = A + B + C$
- b. $X = A.(B + C)$
- c. $X = (A.B) + C$
- d. $X = A + (B.C)$

18. In the Karnaugh map shown in figure 3, which of the loops shown represents a legal grouping?

- a. x
- b. y
- c. z
- d. w

19. Half adder consists of. andGates

- a. EX-OR, AND
- b. EX-OR, OR
- c. EX-OR, NOT
- d. None of the above

20. In half adder EX-OR gate output is

- a. Carry
- b. Remainder
- c. Sum
- d. None of the above

Question 2 (10 points): Write the simplified Boolean expression that is represented by the K-Map shown in figure 3.

Question 3 (20 points): Redraw the circuit shown in figure 2 using NAND gates only.

Figures

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Figure 1

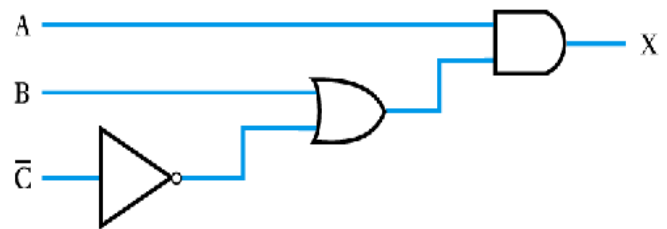


Figure 2

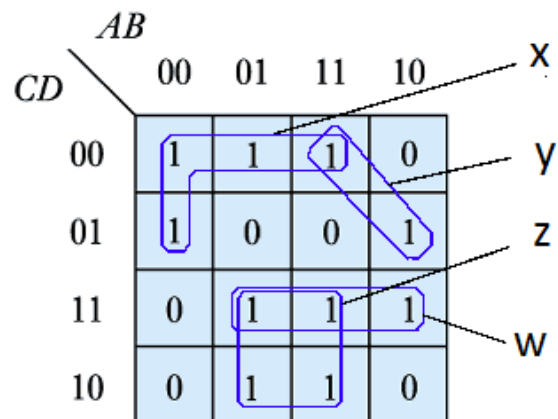


Figure 3